

mageck_nest_comparison_figures

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This report summarises MAGeCK NEST gene-level results across multiple analysis pipelines and visualises score distributions, top-hit overlap, and the relationship between outlier burden and hit strength.

Load results

```
realdeal <- read.delim("data_raw/realdeal_PPI_False_outliers_removal_False.gene_summary.txt")
june <- read.delim("data_raw/june_PPI_False_outliers_removal_False.gene_summary.txt")
noLL <- read.delim("data_raw/noLL_PPI_False_outliers_removal_False.gene_summary.txt")
noLLctrlnorm <- read.delim("data_raw/noLLctrlnorm_PPI_False_outliers_removal_False.gene_summary.txt")

realdeal_outliers <- read.delim("data_raw/realdeal_suggested_outliers_by_gene.txt")
june_outliers <- read.delim("data_raw/june_suggested_outliers_by_gene.txt")
noLL_outliers <- read.delim("data_raw/noLL_suggested_outliers_by_gene.txt")
noLLctrlnorm_outliers <- read.delim("data_raw/noLLctrlnorm_suggested_outliers_by_gene.txt")
```

Define valid genes (that NEST was able to calculate scores for)

```
valid_realdeal <- realdeal %>%
  filter(!is.na(Pos.z) & is.finite(Pos.z),
         !is.na(Neg.z) & is.finite(Neg.z),
         !is.na(CTRL.z) & is.finite(CTRL.z),
         sgRNA >= 3) %>%
  mutate(pipeline = "realdeal")

valid_june <- june %>%
  filter(!is.na(Pos.z) & is.finite(Pos.z),
         !is.na(Neg.z) & is.finite(Neg.z),
         !is.na(CTRL.z) & is.finite(CTRL.z),
         sgRNA >= 3) %>%
  mutate(pipeline = "june")

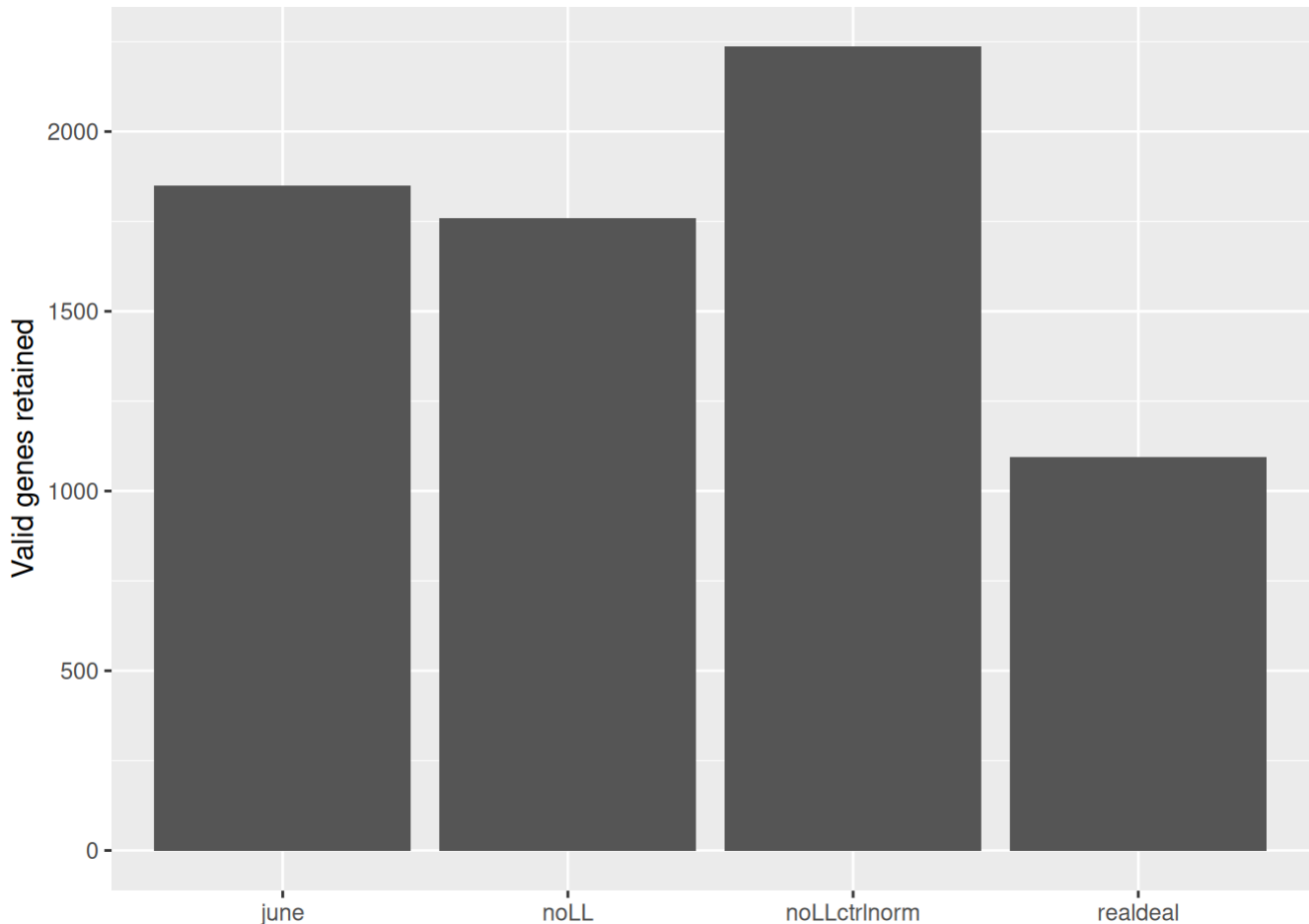
valid_noLL <- noLL %>%
  filter(!is.na(Pos.z) & is.finite(Pos.z),
         !is.na(CTRL.z) & is.finite(CTRL.z),
         sgRNA >= 3) %>%
  mutate(pipeline = "noLL")

valid_noLLctrlnorm <- noLLctrlnorm %>%
  filter(!is.na(Pos.z) & is.finite(Pos.z),
         !is.na(CTRL.z) & is.finite(CTRL.z),
         sgRNA >= 3) %>%
  mutate(pipeline = "noLLctrlnorm")
```

Number of valid genes retained

```
valid_counts <- bind_rows(
  summarise(valid_realdeal, n_valid = n(), pipeline = "realdeal"),
  summarise(valid_june, n_valid = n(), pipeline = "june"),
  summarise(valid_noLL, n_valid = n(), pipeline = "noLL"),
  summarise(valid_noLLctrlnorm, n_valid = n(), pipeline = "noLLctrlnorm") )

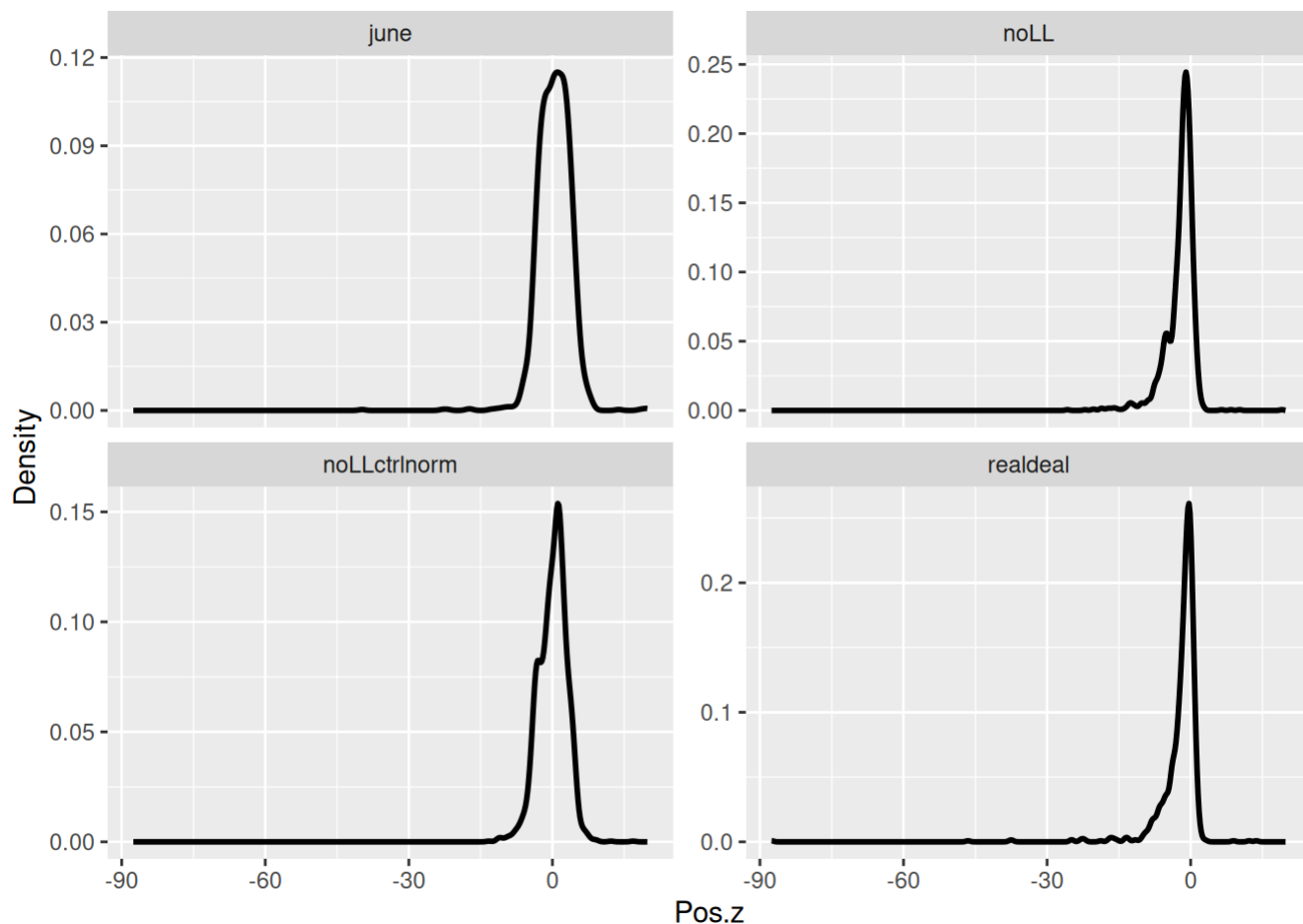
ggplot(valid_counts, aes(x = pipeline, y = n_valid)) + geom_col() + labs(x = NULL, y
= "Valid genes retained")
```



distribution of positive z-scores

```
pos_df <- bind_rows(
  valid_realdeal %>% select(Gene, Pos.z, pipeline),
  valid_june %>% select(Gene, Pos.z, pipeline),
  valid_noLL %>% select(Gene, Pos.z, pipeline),
  valid_noLLctrlnorm %>% select(Gene, Pos.z, pipeline) )

ggplot(pos_df, aes(x = Pos.z)) +
  geom_density(linewidth = 1) +
  facet_wrap(~ pipeline, scales = "free_y") +
  labs(x = "Pos.z", y = "Density")
```

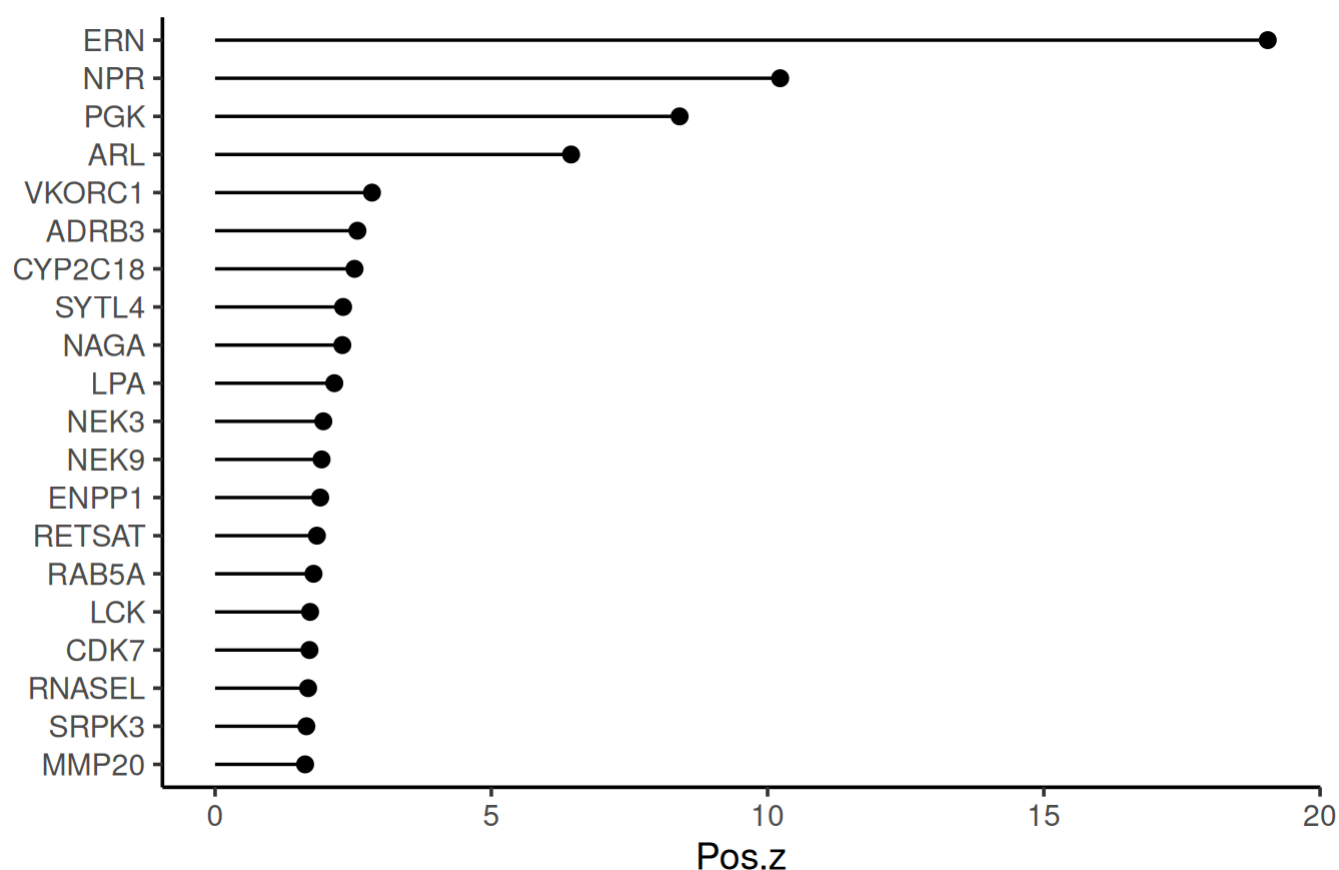


Ranked UL hits (pos.z)

```
top20_noLL <- valid_noLL %>%
  arrange(desc(Pos.z)) %>%
  slice_head(n = 20) %>%
  mutate(Gene = factor(Gene, levels = rev(Gene)))

ggplot(top20_noLL, aes(x = Pos.z, y = Gene)) +
  geom_segment(aes(x = 0, xend = Pos.z, y = Gene, yend = Gene), linewidth = 0.6) +
  geom_point(size = 2.5) +
  labs(
    x = "Pos.z",
    y = NULL,
    title = "Top 20 UL hits after removing LL"
  ) +
  theme_classic(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5))
```

Top 20 UL hits after removing LL



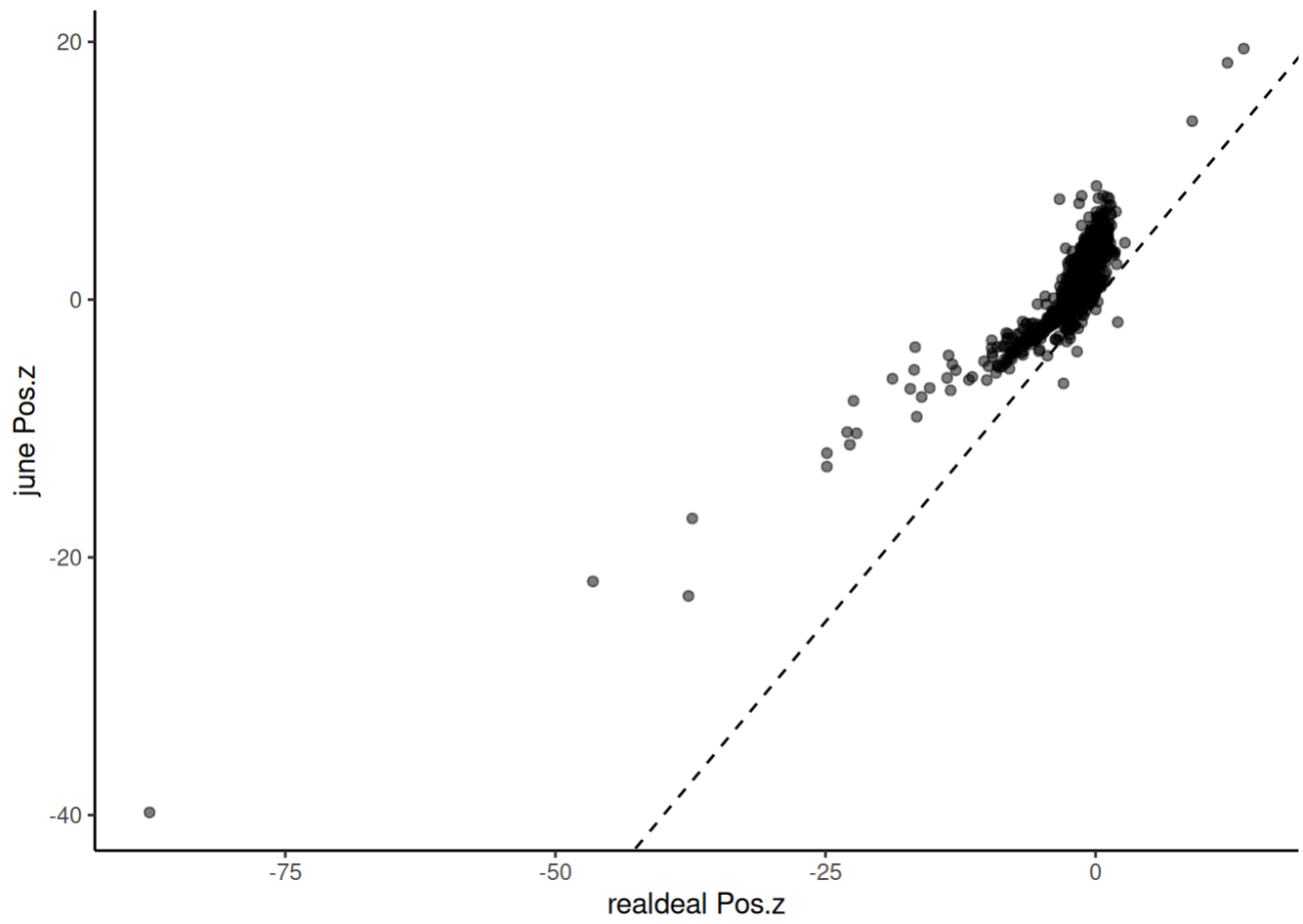
Compare between pipelines

```

realdeal_vs_june <- inner_join(
  valid_realdeal %>% select(Gene, Pos.z_realdeal = Pos.z),
  valid_june %>% select(Gene, Pos.z_june = Pos.z),
  by = "Gene"
)

ggplot(realdeal_vs_june, aes(x = Pos.z_realdeal, y = Pos.z_june)) +
  geom_point(alpha = 0.5) +
  geom_abline(slope = 1, intercept = 0, linetype = 2) +
  labs(
    x = "realdeal Pos.z",
    y = "june Pos.z"
  ) +
  theme_classic()

```



no LL scatter:

```
noLL_vs_ctrl <- inner_join(
  valid_noLL %>% select(Gene, Pos.z_noLL = Pos.z),
  valid_noLLctrlnorm %>% select(Gene, Pos.z_noLLctrlnorm = Pos.z),
  by = "Gene")

top_noLL <- valid_noLL %>%
  arrange(desc(Pos.z)) %>%
  slice_head(n = 20) %>%
  pull(Gene)

top_noLLctrl <- valid_noLLctrlnorm %>%
  arrange(desc(Pos.z)) %>%
  slice_head(n = 20) %>%
  pull(Gene)

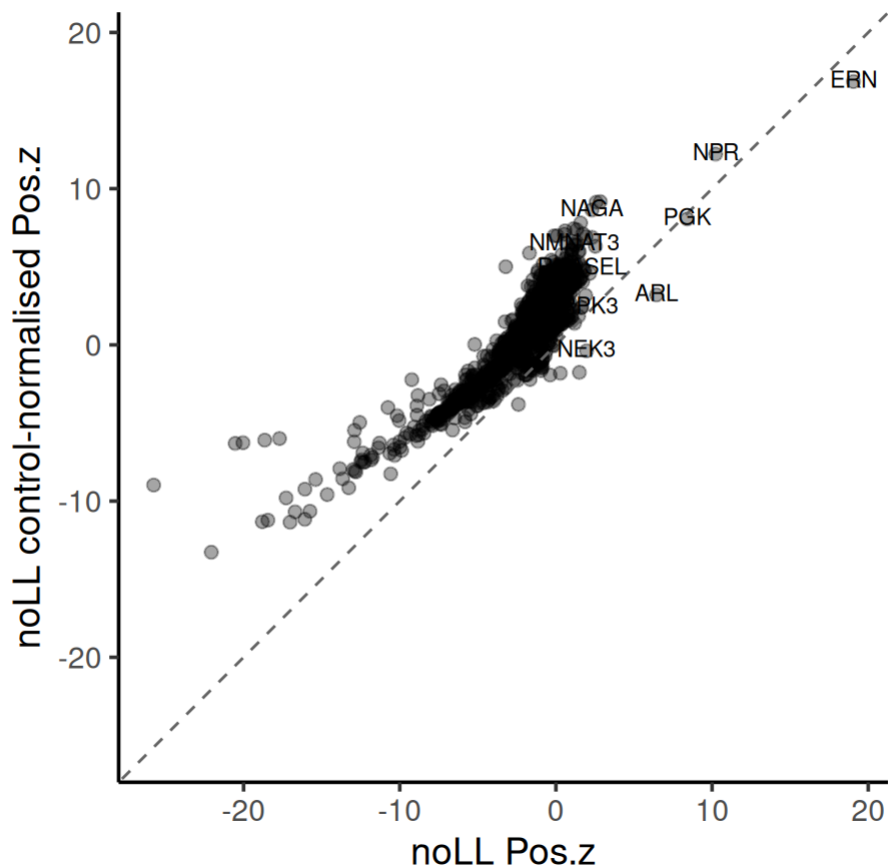
label_genes <- union(top_noLL, top_noLLctrl)

plot_df <- noLL_vs_ctrl %>%
  mutate(to_label = ifelse(Gene %in% label_genes, Gene, NA))

lims <- range(c(plot_df$Pos.z_noLL, plot_df$Pos.z_noLLctrlnorm), na.rm = TRUE)

ggplot(plot_df, aes(x = Pos.z_noLL, y = Pos.z_noLLctrlnorm)) +
  geom_point(alpha = 0.35, size = 2) +
  geom_abline(slope = 1, intercept = 0, linetype = 2, colour = "grey40") +
  geom_text(
    aes(label = to_label),
    na.rm = TRUE,
    size = 3,
    check_overlap = TRUE,
    nudge_y = 0.15
  ) +
  coord_equal(xlim = lims, ylim = lims) +
  labs(
    x = "noLL Pos.z",
    y = "noLL control-normalised Pos.z",
    title = "UL hit scores are more concordant after removing LL"
  ) +
  theme_classic(base_size = 14) +
  theme(plot.title = element_text(face = "bold", hjust = 0.5))
```

UL hit scores are more concordant after removing LL

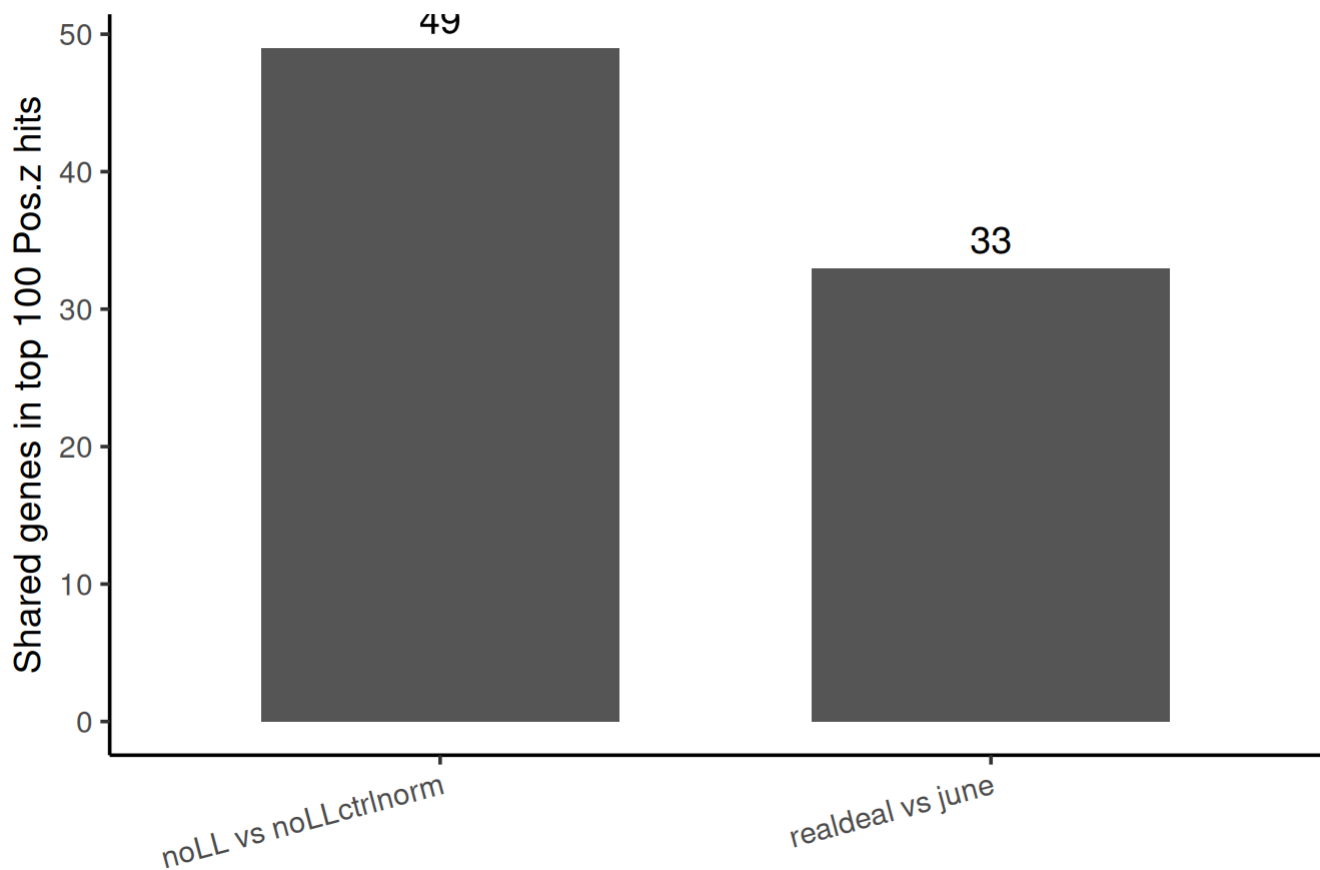


Top 100 overlap

```
overlap_df <- tibble::tibble(
  comparison = c("realdeal vs june", "noLL vs noLLctrlnorm"),
  overlap_top100 = c(33, 49)
)

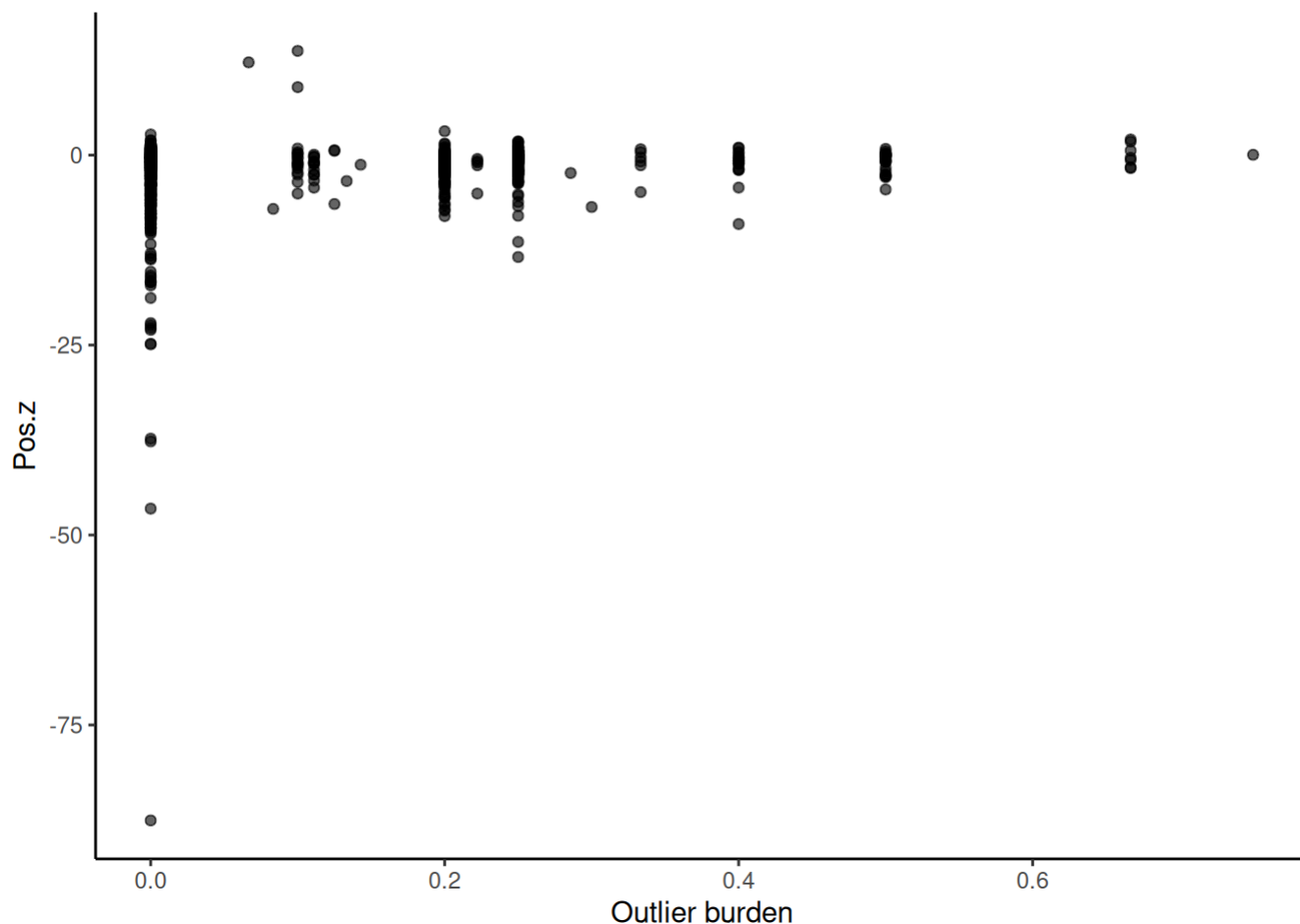
ggplot(overlap_df, aes(x = comparison, y = overlap_top100)) +
  geom_col(width = 0.65) +
  geom_text(aes(label = overlap_top100), vjust = -0.5, size = 5) +
  labs(
    x = NULL,
    y = "Shared genes in top 100 Pos.z hits",
    title = "Top-hit overlap improves after removing LL"
  ) +
  theme_classic(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", hjust = 0.5),
    axis.text.x = element_text(angle = 15, hjust = 1)
  )
```

Top-hit overlap improves after removing LL



Assessing outlier burden among top hits:

```
realdeal_outlier_counts <- realdeal_outliers %>%  
  count(Gene, name = "n_outlier_guides")  
  
realdeal_annot <- valid_realdeal %>%  
  left_join(realdeal_outlier_counts, by = "Gene") %>%  
  mutate(  
    n_outlier_guides = ifelse(is.na(n_outlier_guides), 0, n_outlier_guides),  
    outlier_burden = n_outlier_guides / sgRNA )  
  
ggplot(realdeal_annot, aes(x = outlier_burden, y = Pos.z)) +  
  geom_point(alpha = 0.6) +  
  labs( x = "Outlier burden", y = "Pos.z" ) +  
  theme_classic()
```

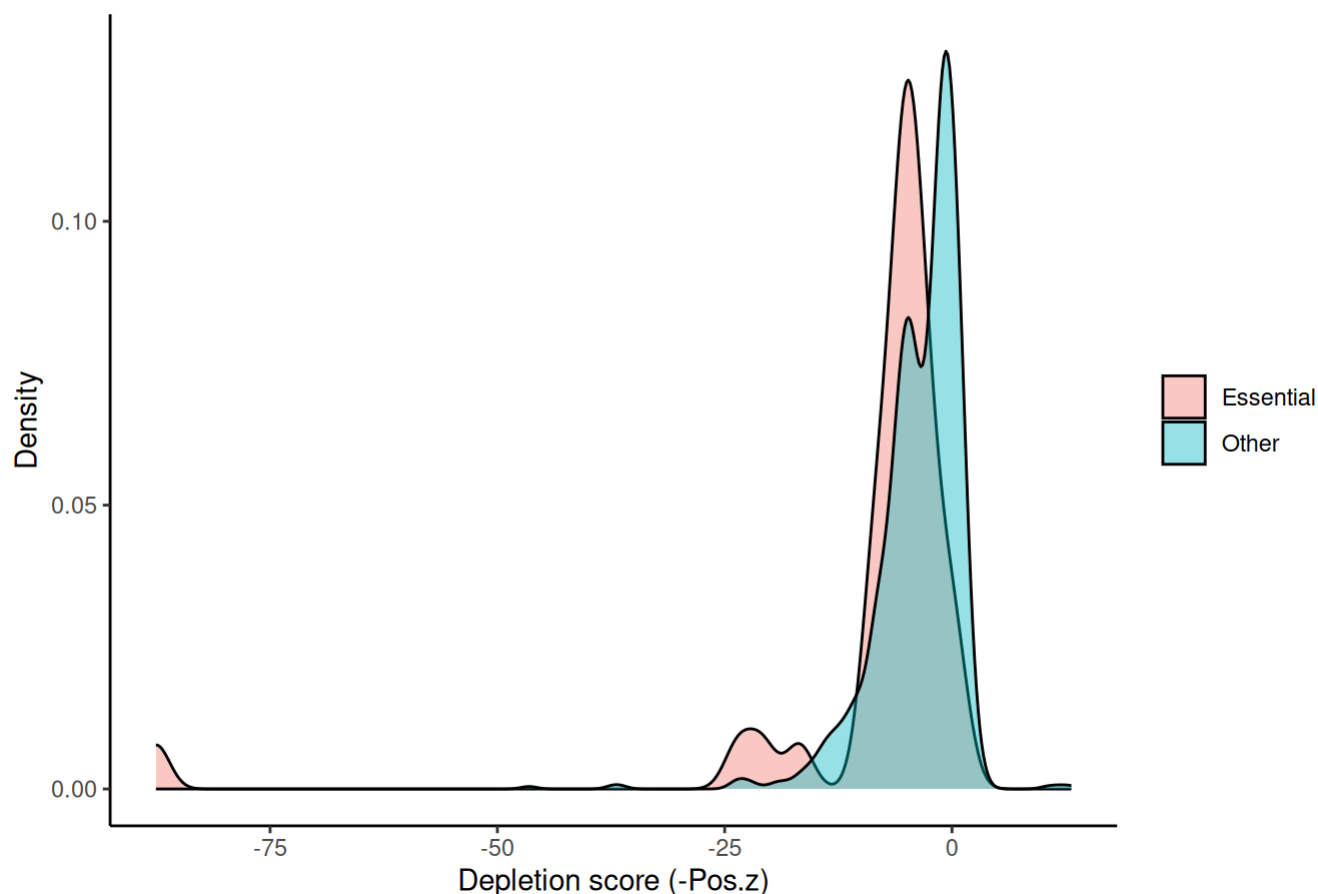



Essential genes are dropping out as expected. The following table and figure uses Gene Set Enrichment Analysis (GSEA) to estimate the dropout of core essential genes (defined by the Johannes Zuber lab and also used in the MAGeCKFlute R package).

```
ess_df <- valid_realdeal %>%
  mutate(
    is_essential = ifelse(Gene %in% core_ess_gene_symbol, "Essential", "Other"),
    depletion = -Pos.z)

ggplot(ess_df, aes(x = Neg.z, fill = is_essential)) +
  geom_density(alpha = 0.4) +
  labs(
    x = "Depletion score (-Pos.z)",
    y = "Density",
    fill = NULL,
    title = "Essential genes show stronger depletion"
  ) +
  theme_classic()
```

Essential genes show stronger depletion



comparing

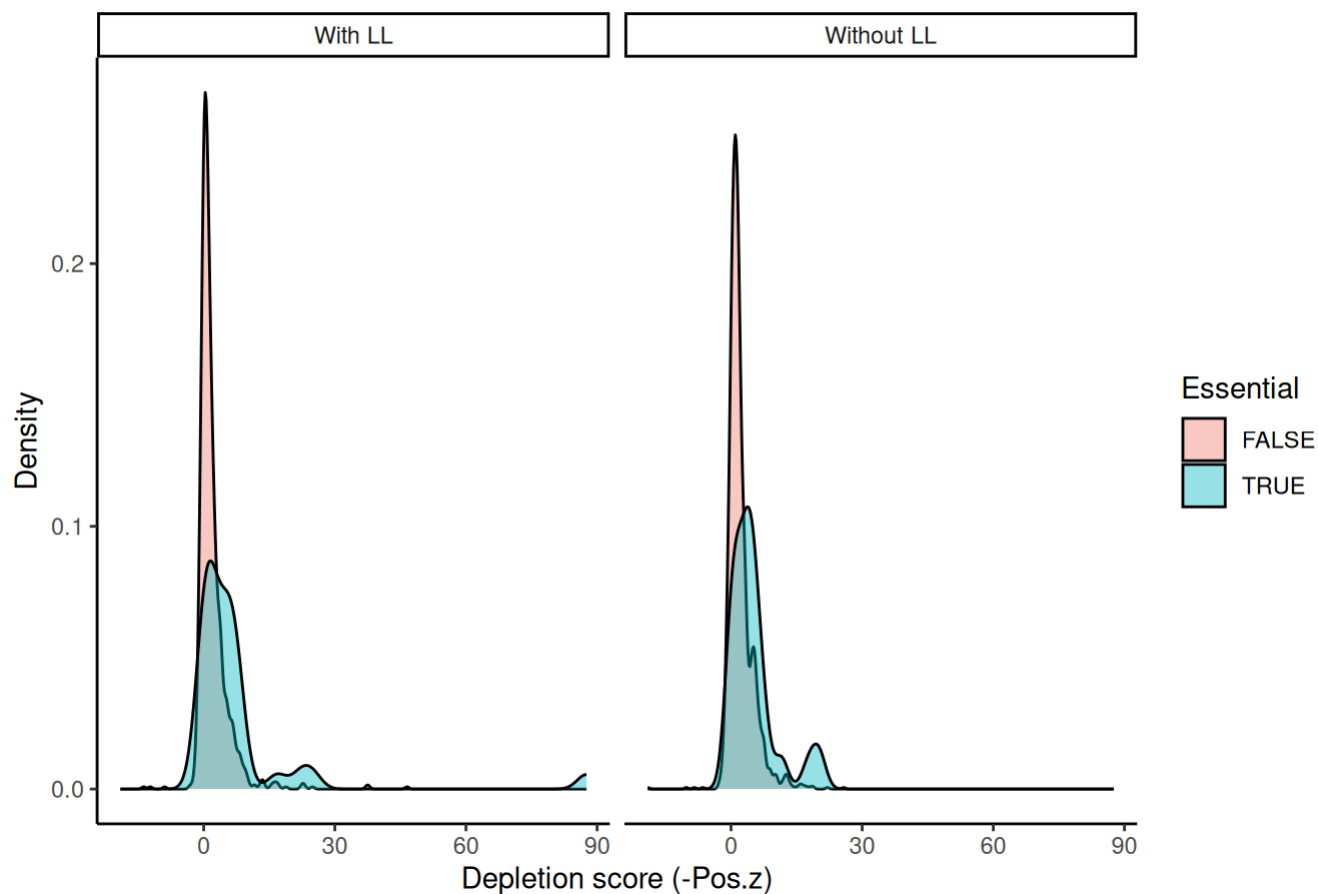
```
ess_realdeal <- valid_realdeal %>%
  mutate(
    is_essential = Gene %in% core_ess_gene_symbol,
    depletion = -Pos.z,
    pipeline = "With LL"
  )

ess_noLL <- valid_noLL %>%
  mutate(
    is_essential = Gene %in% core_ess_gene_symbol,
    depletion = -Pos.z,
    pipeline = "Without LL"
  )

ess_combined <- bind_rows(ess_realdeal, ess_noLL)

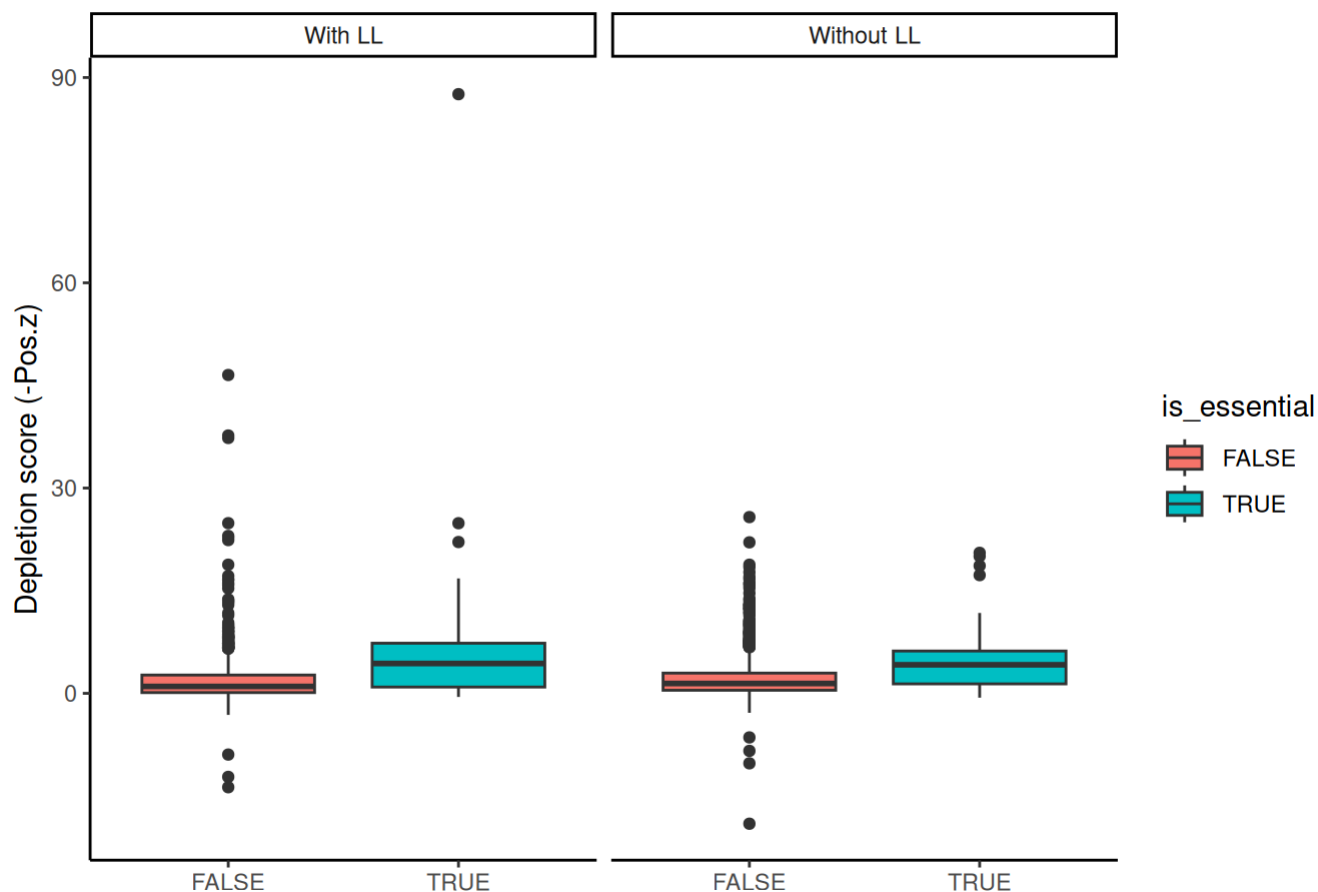
ggplot(ess_combined, aes(x = depletion, fill = is_essential)) +
  geom_density(alpha = 0.4) +
  facet_wrap(~ pipeline) +
  labs(
    x = "Depletion score (-Pos.z)",
    y = "Density",
    fill = "Essential",
    title = "Essential gene depletion improves after removing LL"
  ) +
  theme_classic()
```

Essential gene depletion improves after removing LL



```
ggplot(ess_combined, aes(x = is_essential, y = depletion, fill = is_essential)) +  
  geom_boxplot() +  
  facet_wrap(~ pipeline) +  
  labs(  
    x = NULL,  
    y = "Depletion score (-Pos.z)",  
    title = "Essential genes show stronger depletion after removing LL"  
  ) +  
  theme_classic()
```

Essential genes show stronger depletion after removing LL



```
# With LL
ranks_realdeal <- -valid_realdeal$Pos.z
names(ranks_realdeal) <- valid_realdeal$Gene
ranks_realdeal <- sort(ranks_realdeal, decreasing = TRUE)

# Without LL
ranks_noLL <- -valid_noLL$Pos.z
names(ranks_noLL) <- valid_noLL$Gene
ranks_noLL <- sort(ranks_noLL, decreasing = TRUE)
```

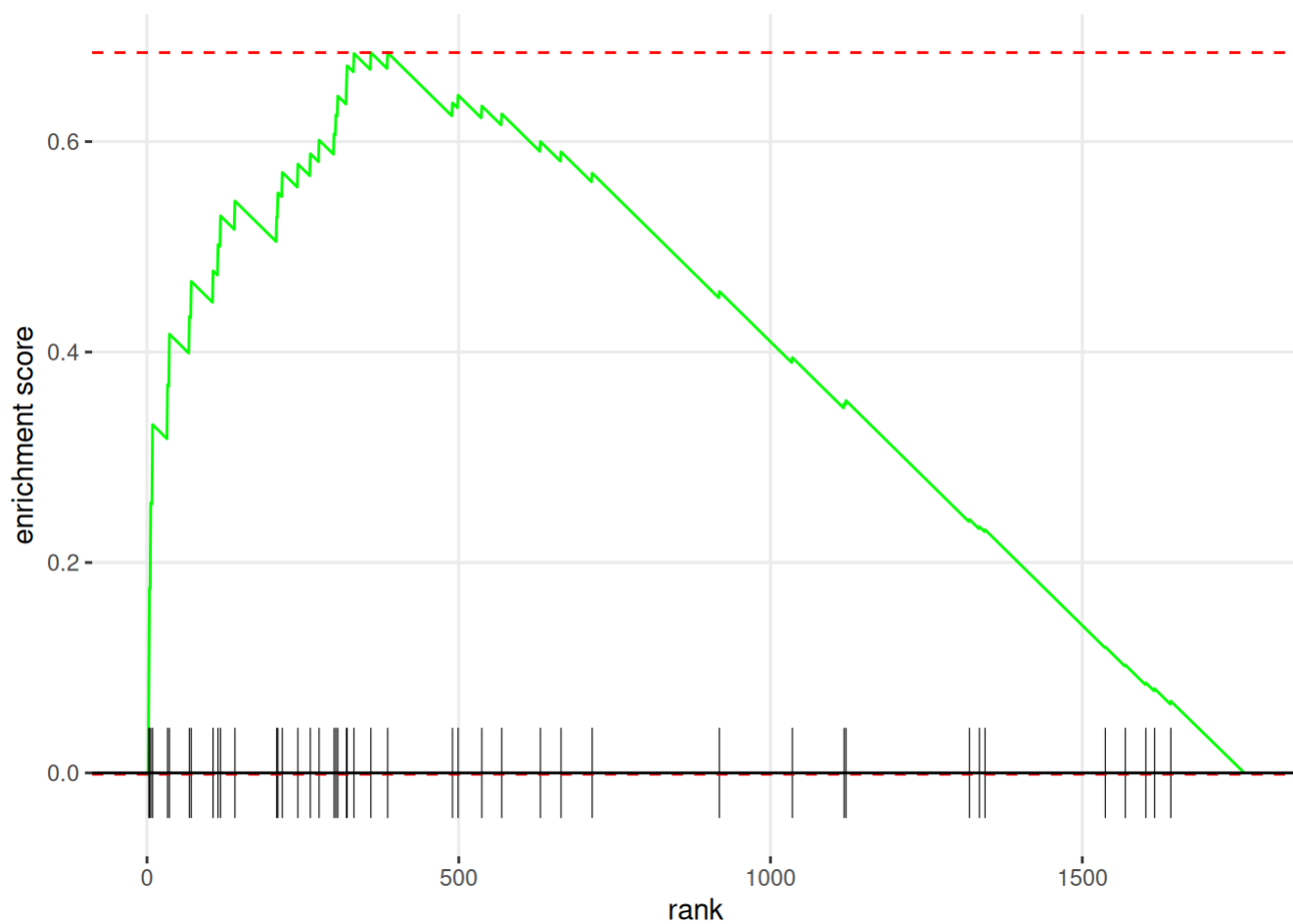
and

```
par(mfrow = c(1,2))

plotEnrichment(core_ess_gene_symbol, ranks_realdeal)
```



```
plotEnrichment(core_ess_gene_symbol, ranks_noLL)
```



boxplot

```
ess_df <- valid_noLL %>%  
  mutate(  
    is_essential = ifelse(Gene %in% core_ess_gene_symbol, "Essential", "Other"),  
    depletion = -Pos.z  
  )  
  
ggplot(ess_df, aes(x = is_essential, y = depletion)) +  
  geom_boxplot() +  
  labs(  
    x = NULL,  
    y = "Depletion score (-Pos.z)",  
    title = "Essential genes show stronger depletion"  
  ) +  
  theme_classic()
```

